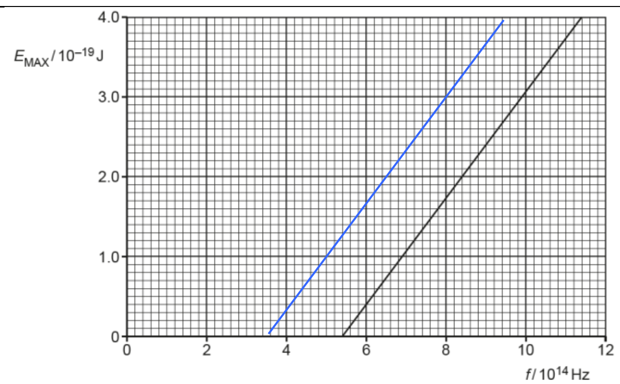
Qn	Suggested Solutions
6(a)	Any two of the following three:
	1. Existence of Threshold Frequency
	Below a certain frequency of the incident radiation, there is no value of maximum kinetic energy, hence no electrons are emitted.
	Energy of radiation is quantised, and if lower than the work function, no electrons will be emitted.
	2. Stopping potential is dependent on frequency but not on intensity.
	For the same intensity of incident light, the higher-frequency light has a larger stopping potential.
	Energy of radiation is quantised and increases so maximum kinetic energy of photoelectrons is higher.
	OR
	For the same frequency of incident light, an increase in the intensity causes the photocurrent to increase but does not affect the stopping potential.
	Energy of radiation is quantised and for same frequency of light, the maximum kinetic energy of photoelectrons does not change.
	3. No Time Lag
	Emission of photoelectrons take place instantly/without a time lag.
	One electron absorbs one photon, with the energy transfer taking place in an instant.
(b)(i)	$hf = hf_o + \text{KE}_{\text{max}}$ $\text{KE}_{\text{max}} = hf - hf_o$ gradient $= h$ $h = \dfrac{(4.00-0)\times 10^{-19}}{(11.4-5.4)\times 10^{14}}$
	$h = 6.67\times 10^{-34}$ J s
(ii)	$\Phi = hf_o$ $= (6.667\times 10^{-34})(5.4\times 10^{14})$
	$\Phi = \dfrac{(3.6\times 10^{-19})}{(1.6\times 10^{-19})}$ eV $= 2.25$ eV
(c)	 The gradient remains the same.
	Since the work function is lower, the threshold frequency increases, the graph is shifted to the left.